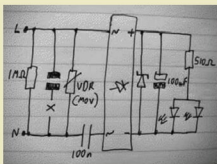




Fig. 2: Yet another model (L) and its circuit (R)



the circuit (in parallel with mains) in order to counter the PF degradation effect (lagging PF, wherein current lags the voltage waveform) of a given inductive load.

Design criteria for varying loads

In most households and industrial establishments, the load may change at unspecified times. Furthermore only the inductive loads (that too if not already compensated for PF inside the equipment itself) will need to be compensated. Hence the PF-compensating circuit should have the intelligence to measure the inductive component of current and bring into the circuit the calculated value of PF capacitors only. If you switch in a PF capacitor even when there is no inductive load, it will result in degradation of the power factor (by introducing a leading PF, wherein the current leads the voltage waveform). A leading or lagging power factor is equally bad.

Manufacturers of repute

Hardly any reputed firm in India manufactures the 'home power saver' devices, although suitable type of PF capacitors for domestic load (small-size, low-loss and high kVAR rating) are available from reputed companies; e.g., EPCOS HomeCap capacitors for home appliances (see the box for specifications). Industrial Automatic Power Factor Correction (APFC) panels are manufactured in India by

many reputed companies like EPIC Energy (Mumbai), MELCON, Alwar (Rajasthan), etc.

Is it legal to use power savers?

To be legal for use in homes, an electrical appliance must conform to House-Hold Electrical Appliances (Quality Control) Order 1981 notified on Nov. 12, 1981 by the central government under Section 3 of the Essential Commodities Act, 1955. There is presently no Indian standard specified for power saver devices, and they don't fall within the purview of Bureau of Indian Standards (BIS) certification. However, the manufacturer of these items is required to be certified by an appropriate authority designated under the Act (for NCT of Delhi, it is Commissioner of Industries), after ensuring the quality and safety aspects of the item.

What is inside the power saver?

Most of the power saver devices that are available in the market under

EPCOS HomeCap specifications

Dielectric	: Polypropylene film
Rated voltage	: 400V (application voltage 127V to 400V)
Capacitance	: 5 to 33 μ F
Reactive power	: 0.25 to 1.66 kVAR
Frequency	: 50/60 Hz
Diameter	: 40 mm
Height	: 70 to 105 mm

various names (Arven Power Saver, A2-Power Saver-250, EnergySmart Power Planner, Watt Wizard, etc) may be just an over-priced PF capacitor, as its main element. The other elements may be transient voltage suppressors/voltage-dependant resistors/metal-oxide varistors, RF bypass capacitors, transformers for differential-mode and common-mode noise rejection, diodes, LEDs and resistors. Expensive ones may

Anatomy of a typical power saver

The website 'www.bigclive.com/power.htm' starts with the advice "Just plug one of these amazing devices into any socket in your house and save up to 35 per cent on your electricity bill. It sounds too good to be true. Let's take one to bits so we can copy it and save lots of money." The physical appearance, inside contents and the circuit are shown in Figs 1 and 2.

A similar incidence related to Arven power saver, purportedly tested by NABL-accredited Lab at Kaakalur in Tamil Nadu, was brought to the notice of EFY by Vinay Chaddha, managing director of GVC Systems, New Delhi. Its analysis was also published under 'Q&A' section of Sept. 2007 issue of EFY. The manufacturer, when contacted, evaded production of its test report from the NABL.

An actual test report

Fig. 3 shows the extract of a power saver test report issued by NABL-accredited ERTL (North) for a similar product referred to as an 'ideal power saver' by its manufacturer. This test report clearly indicates that if you have a fridge and air-conditioner (probably not having high BEE Energy Star ratings) 'on' with some light load, your